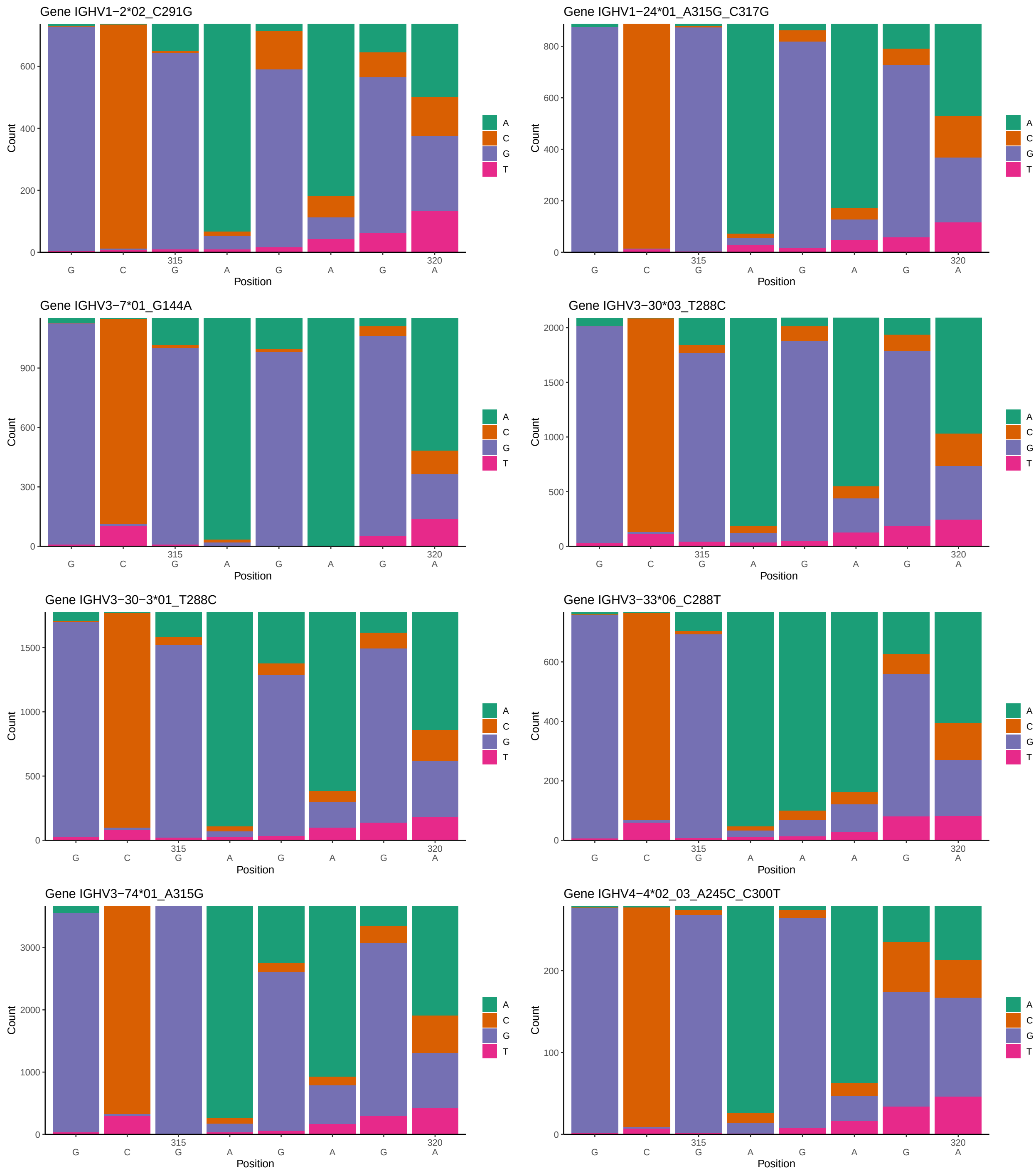
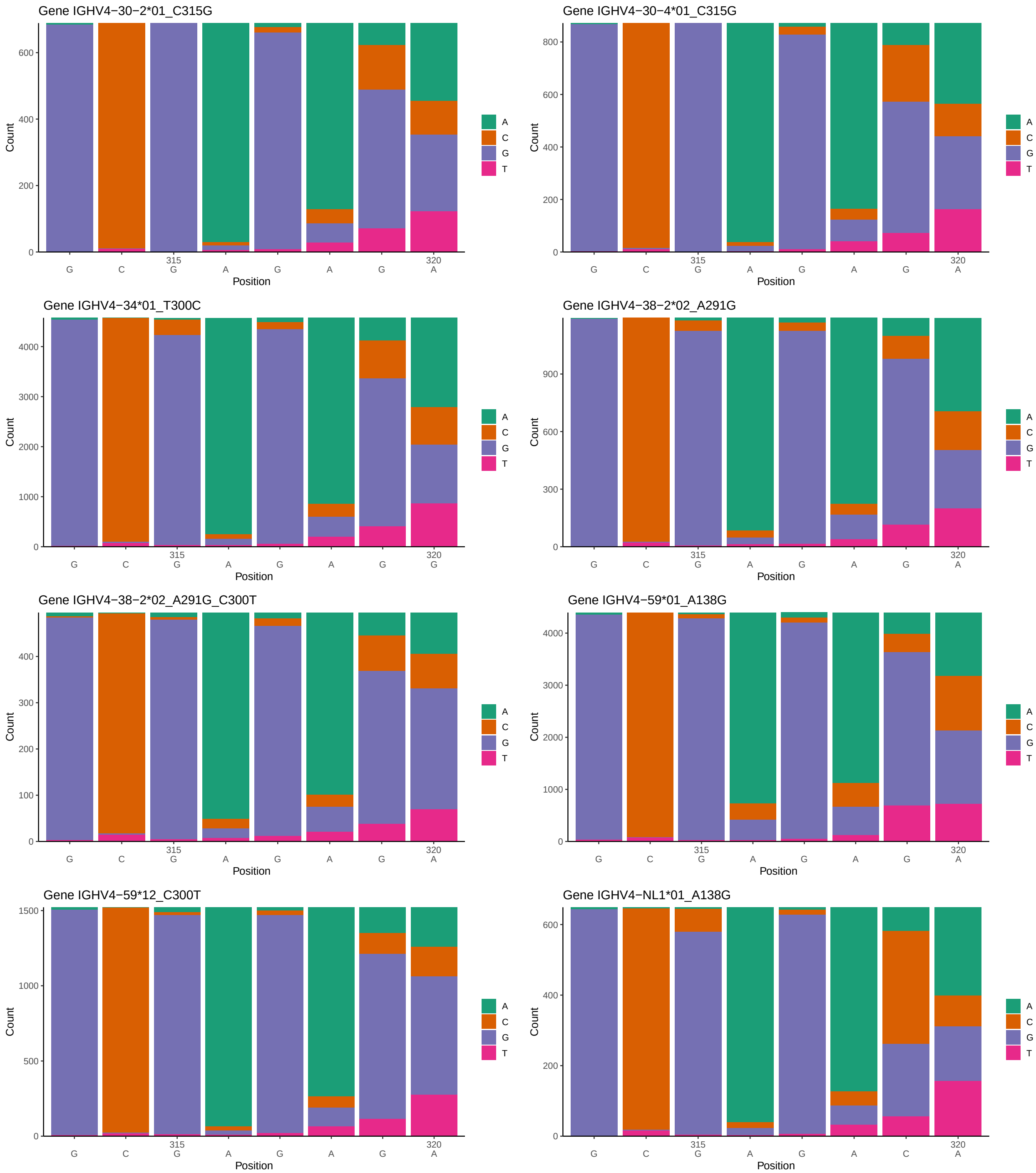
# OGRDBstats Report

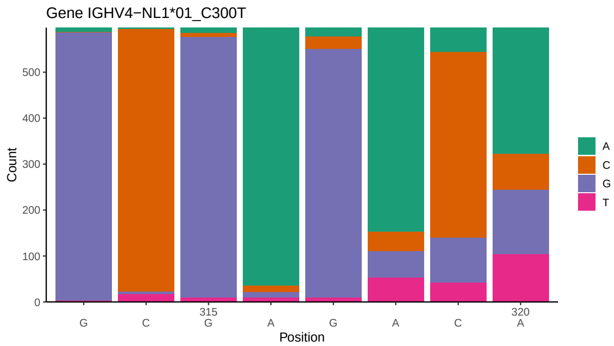
## Contents

# 1 Novel sequence analysis

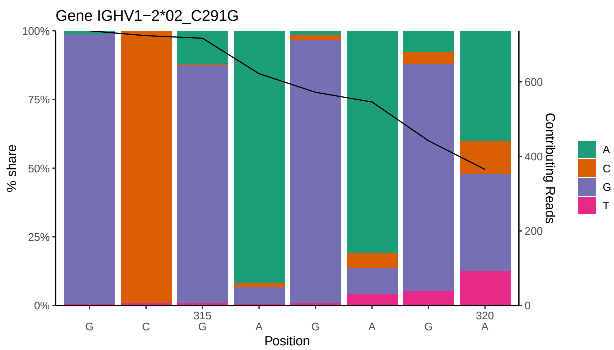
## 1.1 End-nucleotide composition

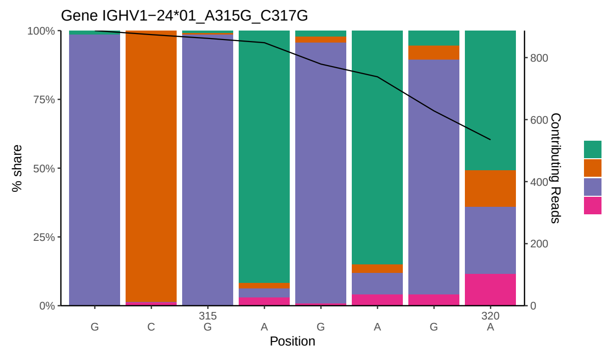


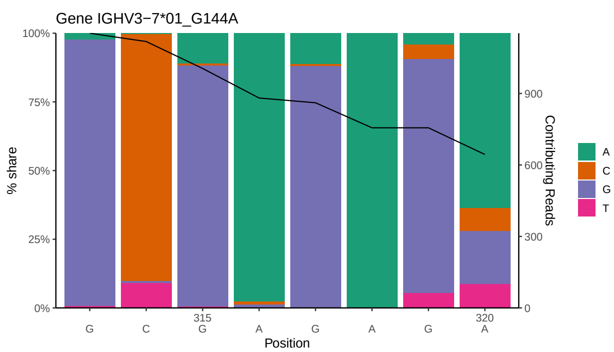

Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*02_A291G
Gene IGHV4−38−2*02_A291G_C300T
Gene IGHV4−59*01_A138G
Gene IGHV4−59*12_C300T
Gene IGHV4−NL1*01_A138G
Count
Position
315
320
A
C
G
T

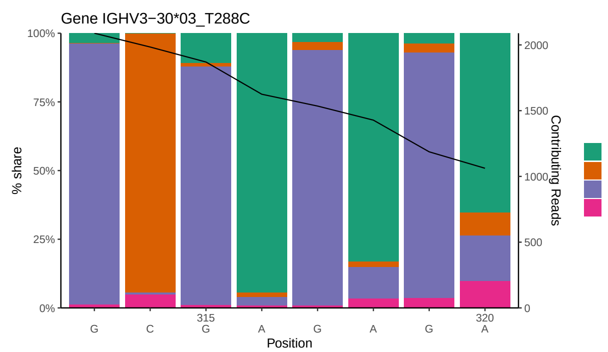


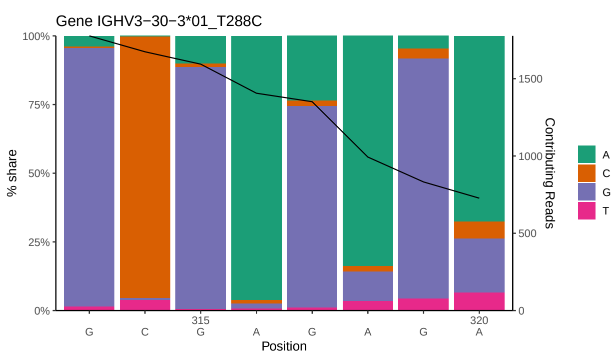

## 1.2 Per-nucleotide consensus where previous nucleotides match the consensus

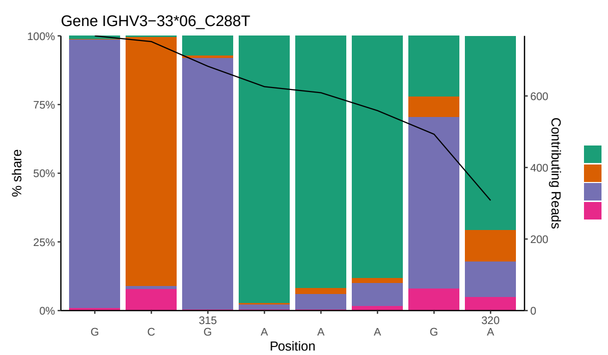


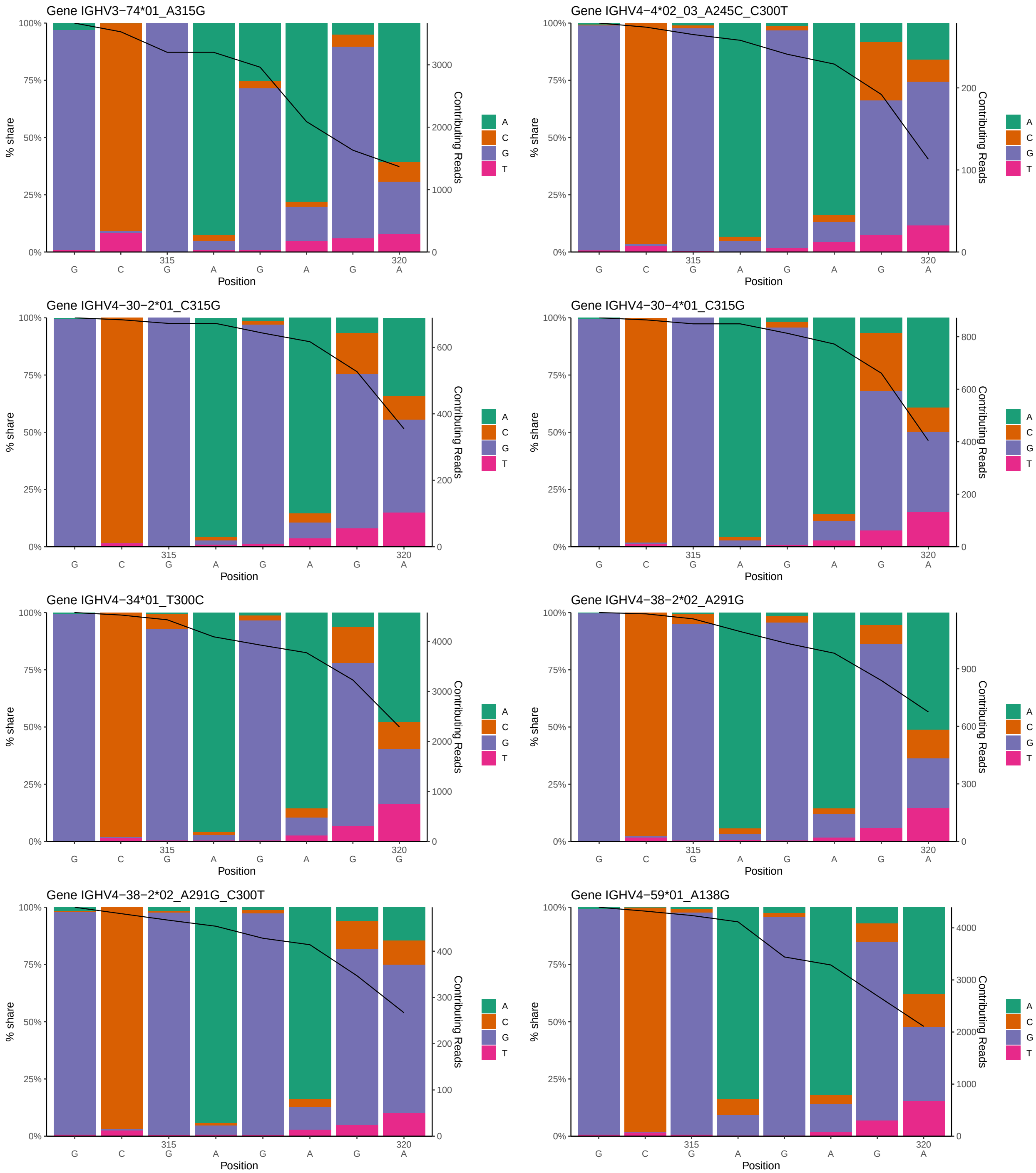



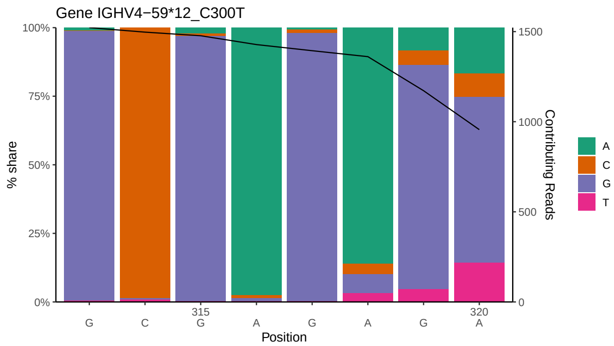



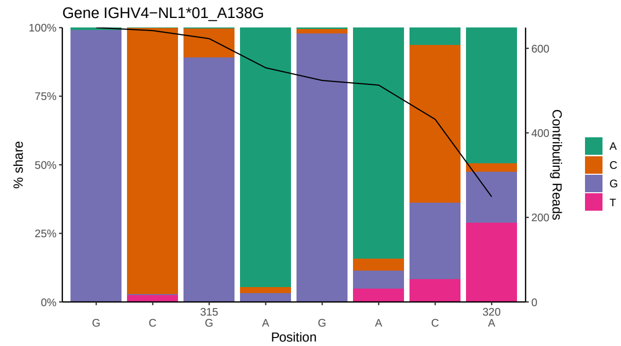



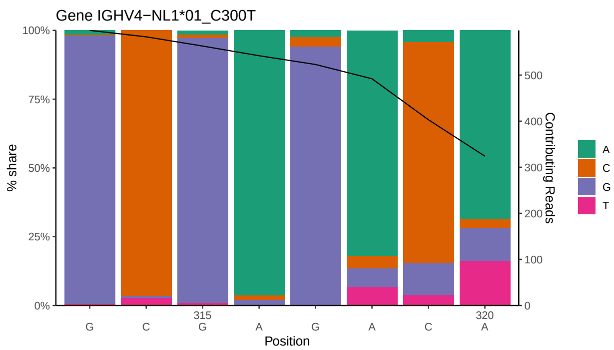



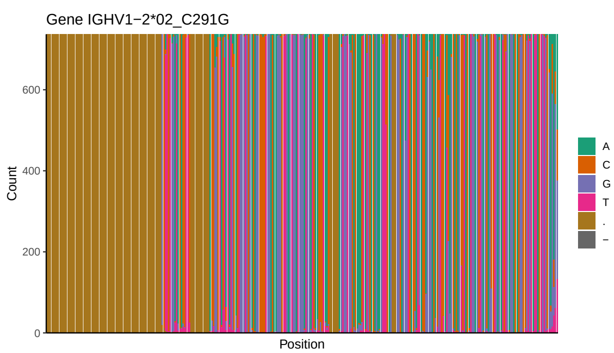

Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*02_A291G
Gene IGHV4−38−2*02_A291G_C300T
Gene IGHV4−59*01_A138G
% share
Contributing Reads
Position
A
C
G
T

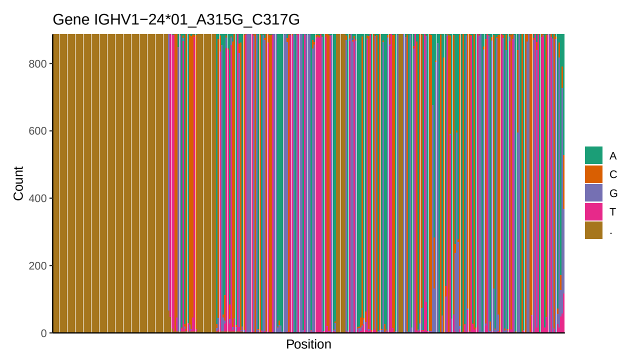


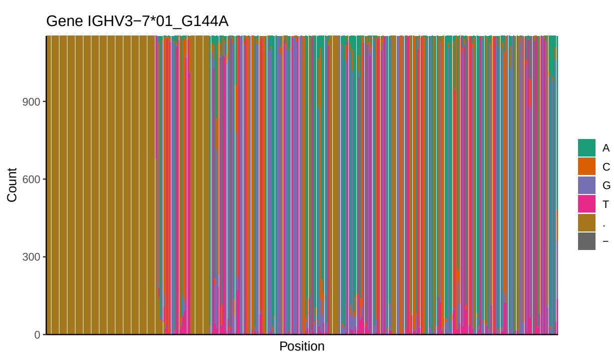



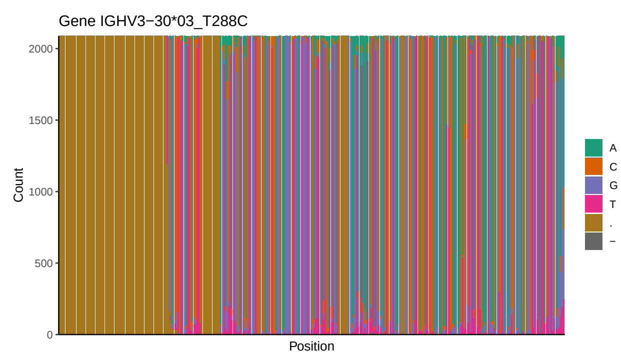



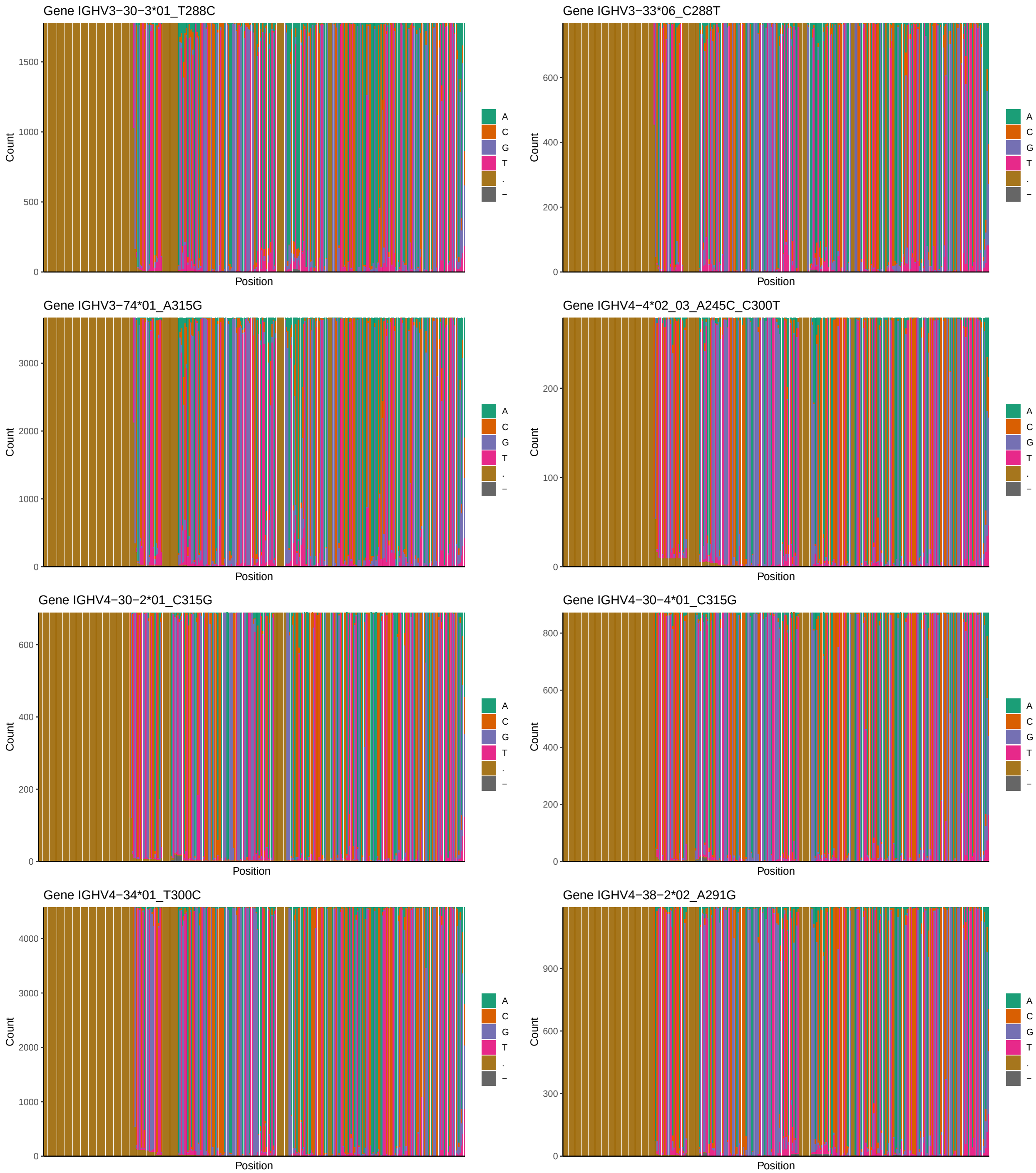

## 1.3 Whole-sequence composition of each assigned read

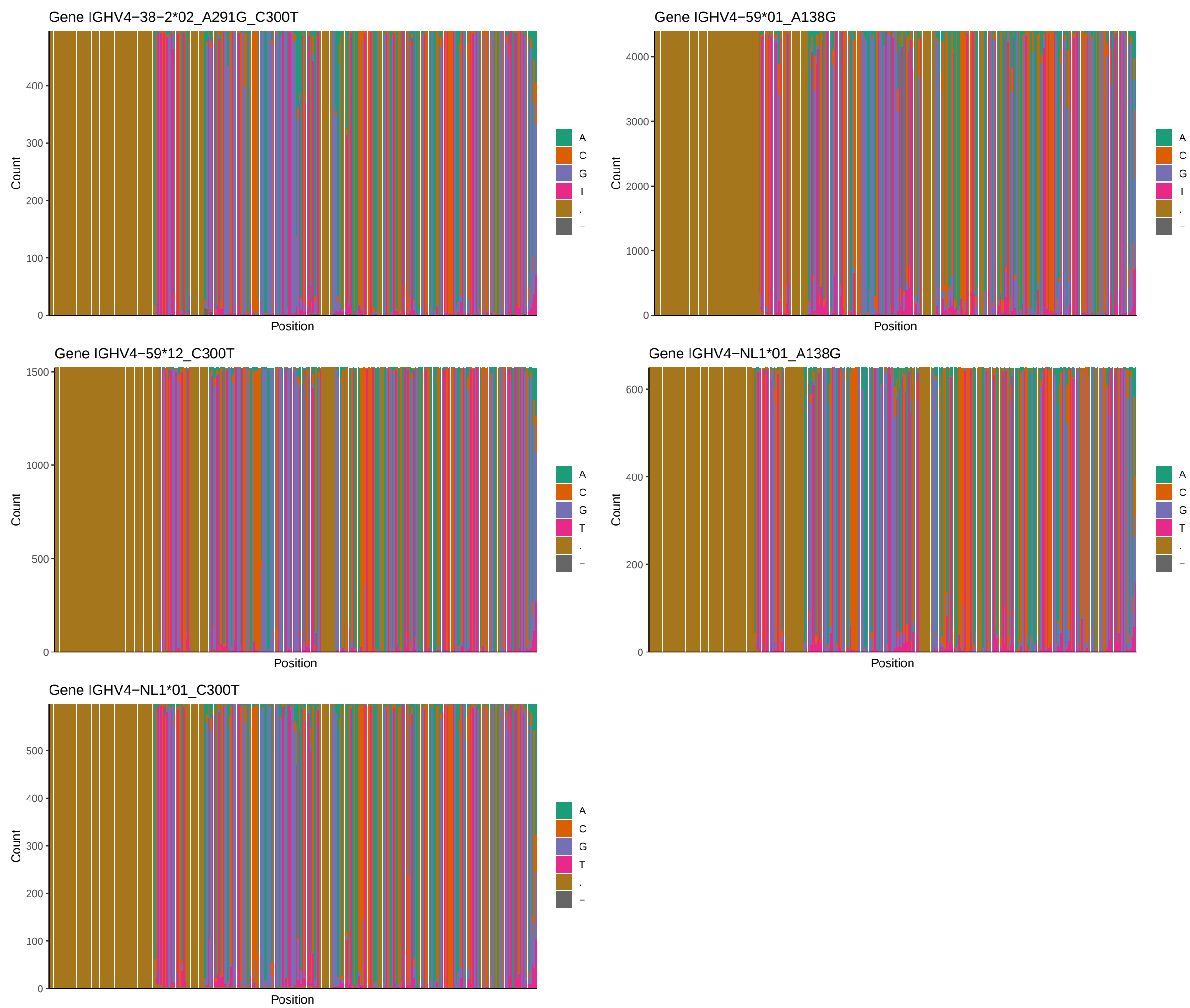


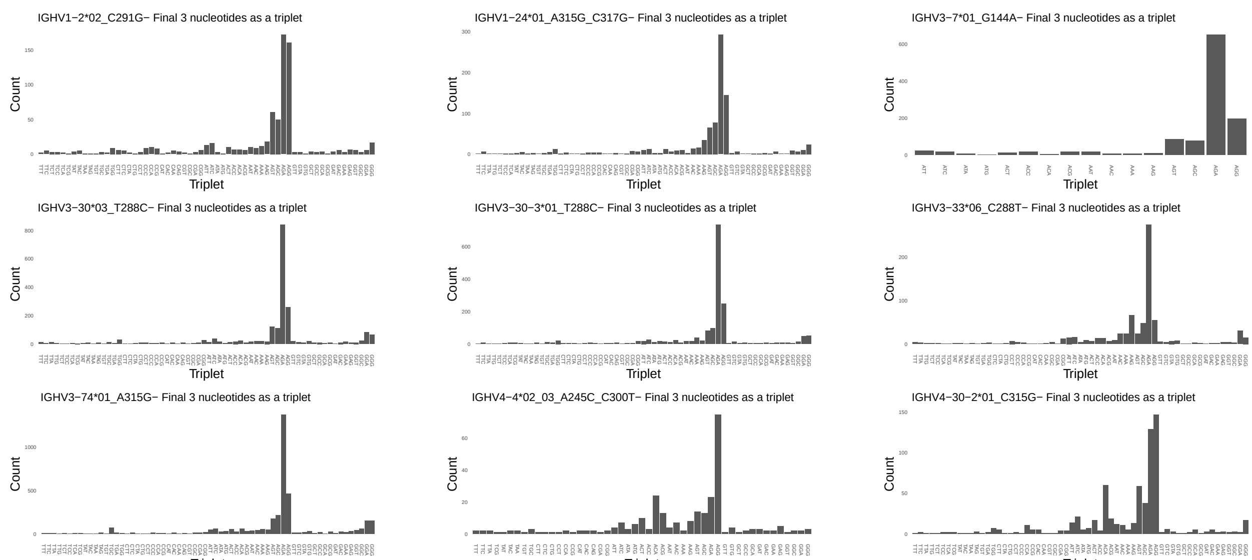



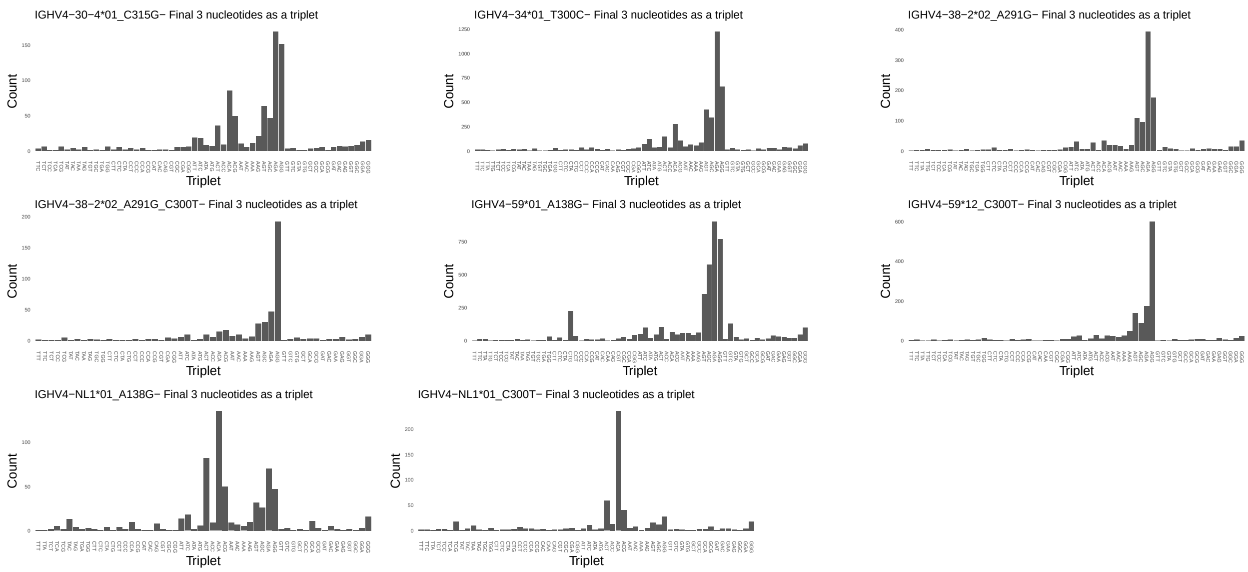




Gene IGHV3−30−3*01_T288C
Gene IGHV3−33*06_C288T
Gene IGHV3−74*01_A315G
Gene IGHV4−4*02_03_A245C_C300T
Gene IGHV4−30−2*01_C315G
Gene IGHV4−30−4*01_C315G
Gene IGHV4−34*01_T300C
Gene IGHV4−38−2*02_A291G
Count
Position
A
C
G
T
.
−

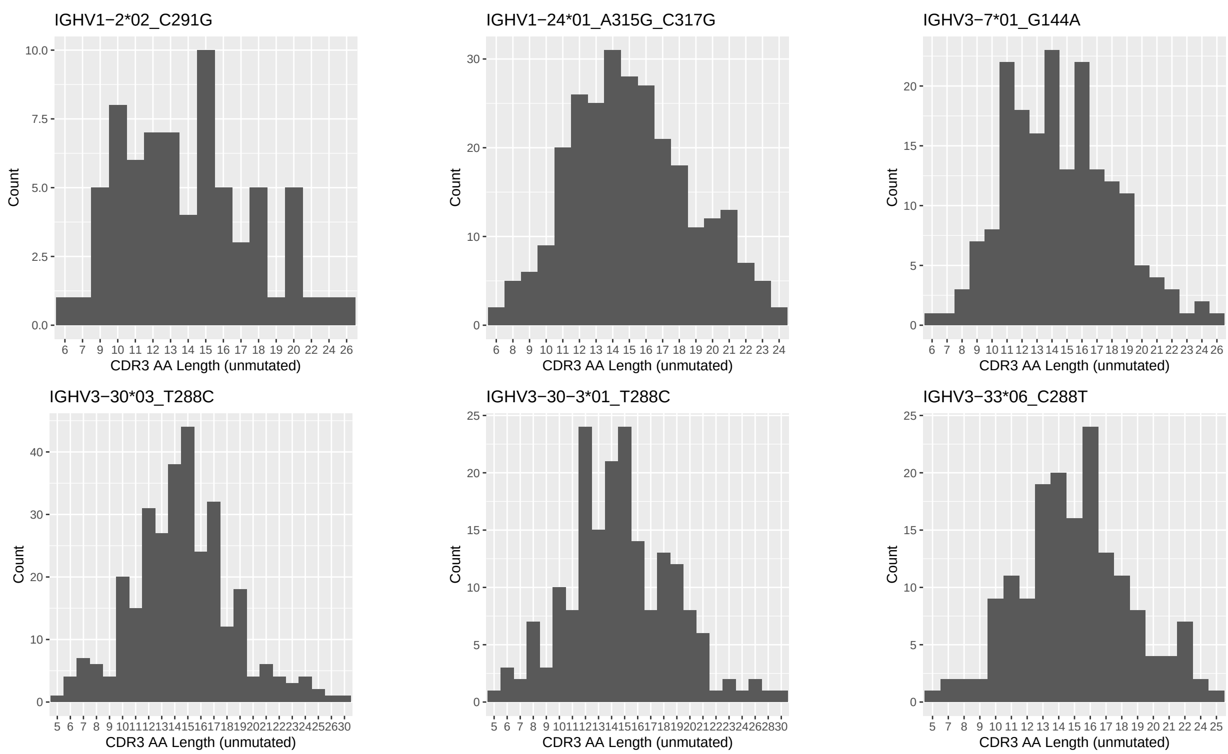


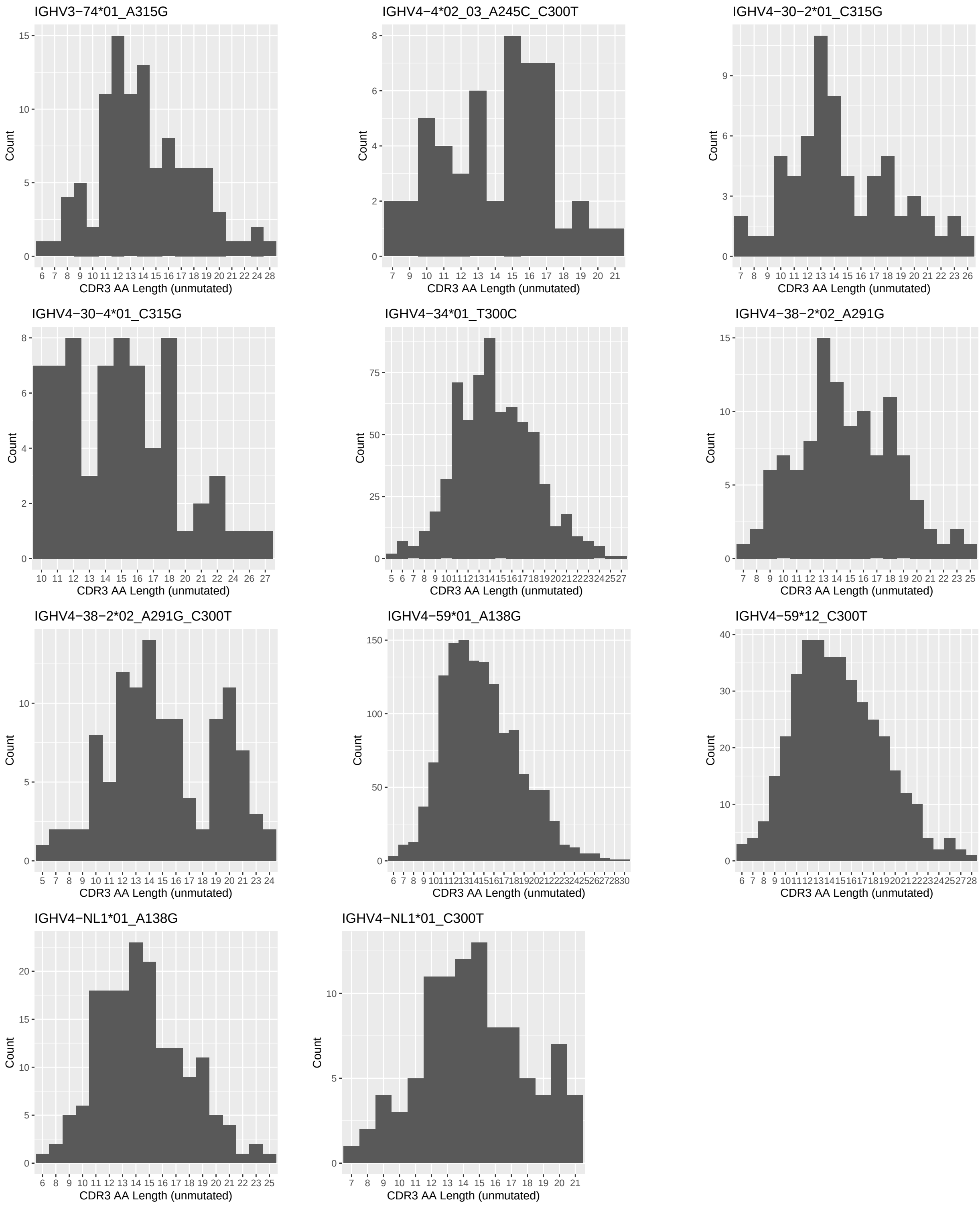

## 1.4 Final three nucleotides: frequency of each observed triplet



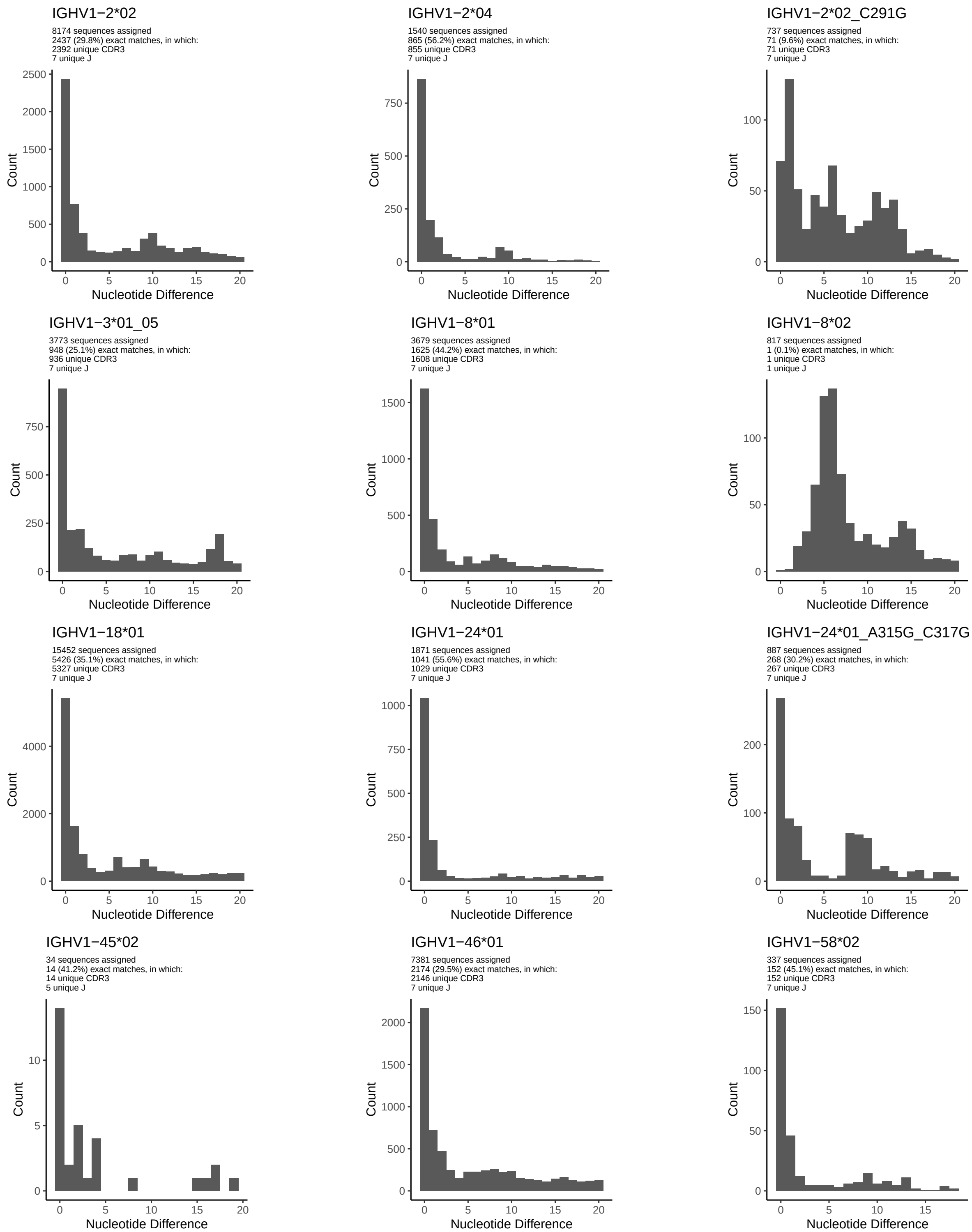


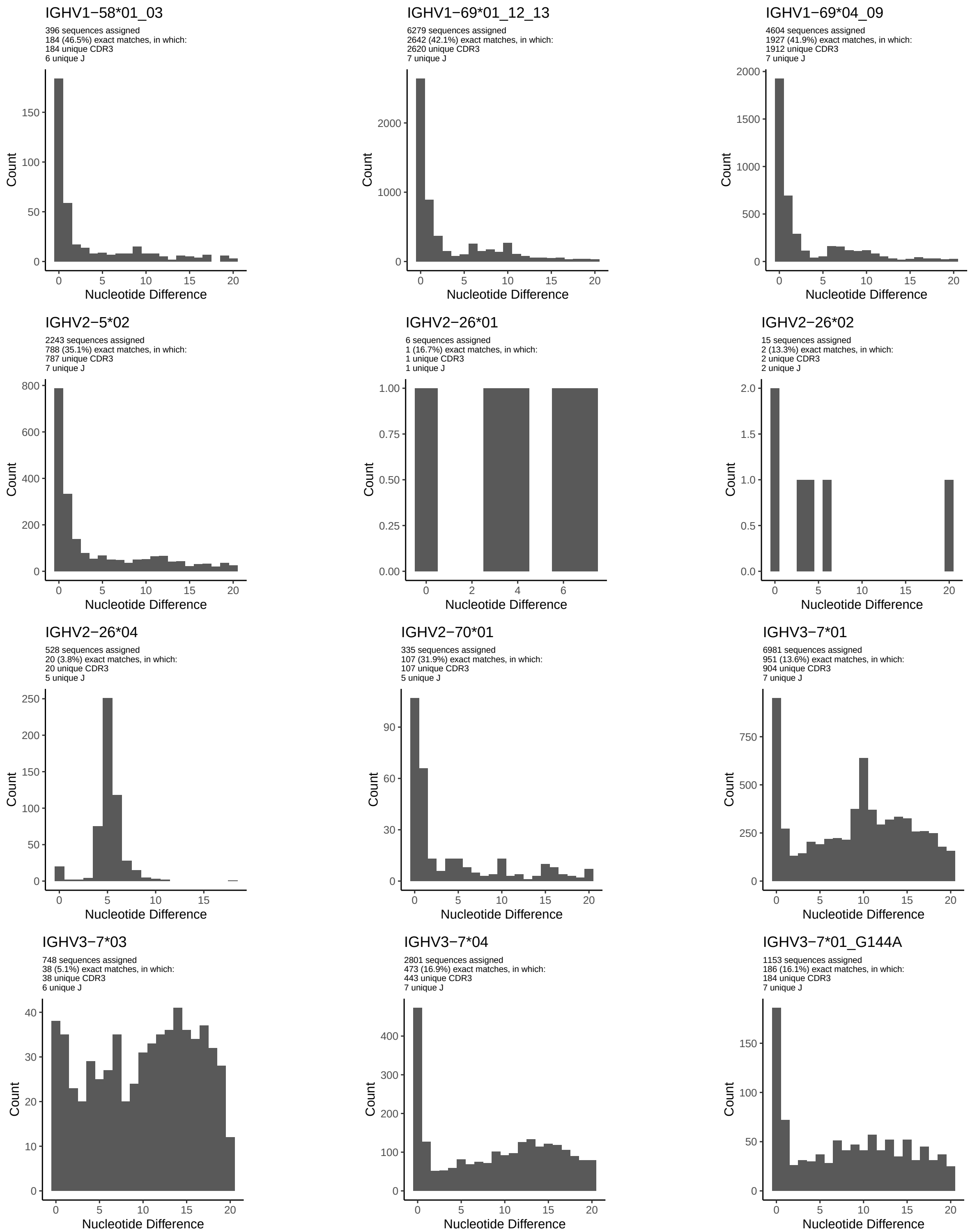

## 1.5 CDR3 length distribution, in assignments to novel alleles

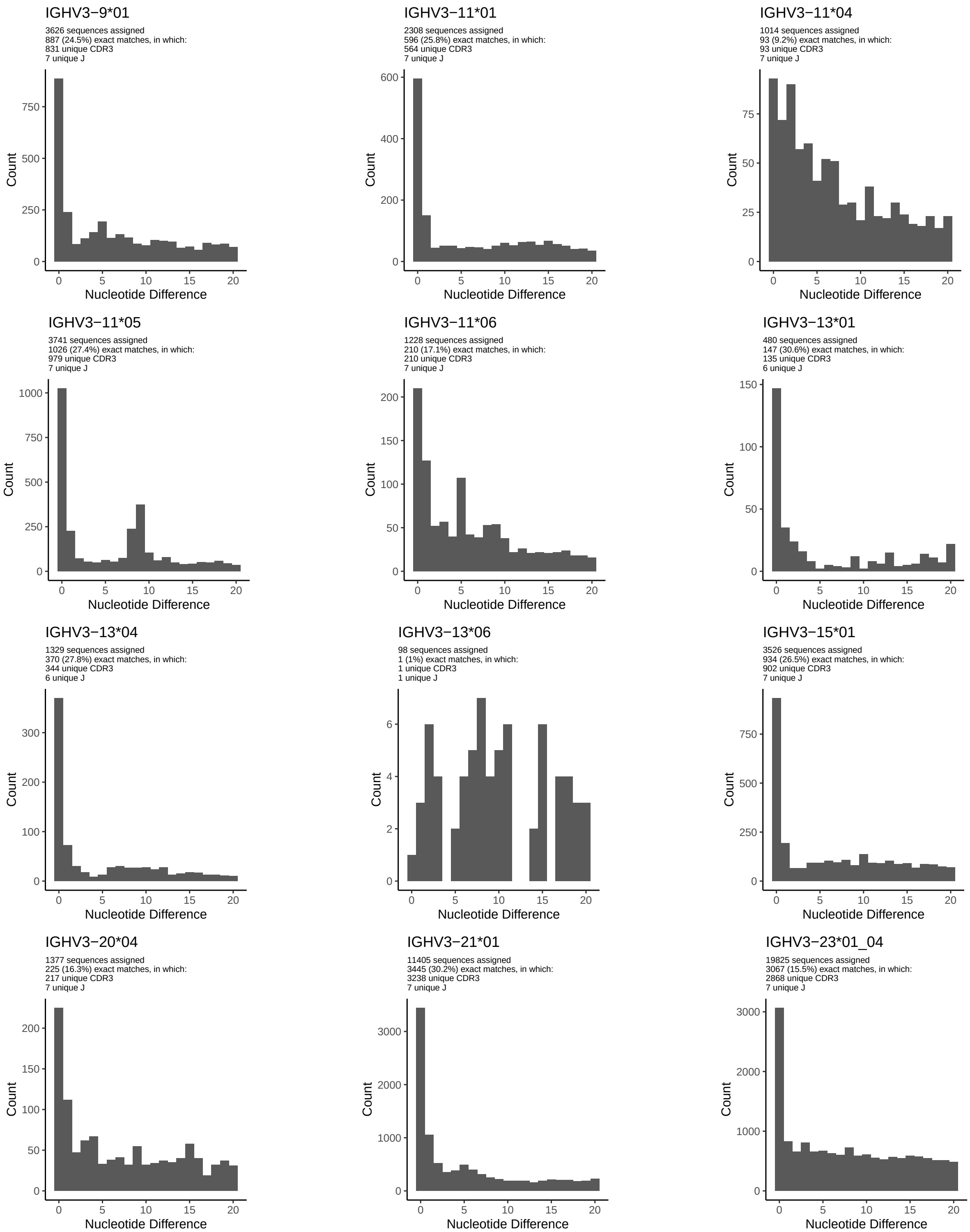


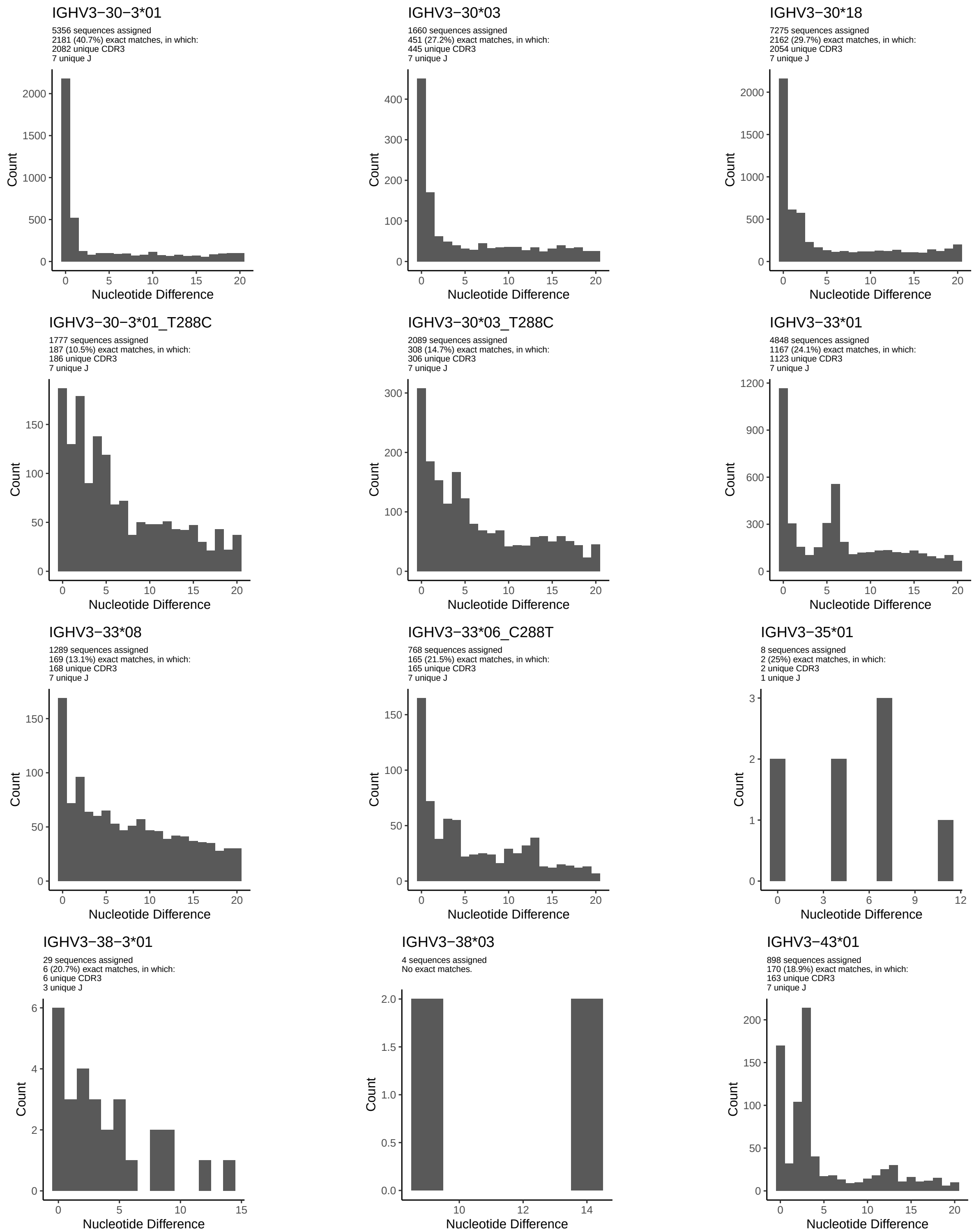

IGHV3−74*01_A315G
IGHV4−4*02_03_A245C_C300T
IGHV4−30−2*01_C315G
IGHV4−30−4*01_C315G
IGHV4−34*01_T300C
IGHV4−38−2*02_A291G
IGHV4−38−2*02_A291G_C300T
IGHV4−59*01_A138G
IGHV4−59*12_C300T
IGHV4−NL1*01_A138G
IGHV4−NL1*01_C300T
Count
CDR3 AA Length (unmutated)

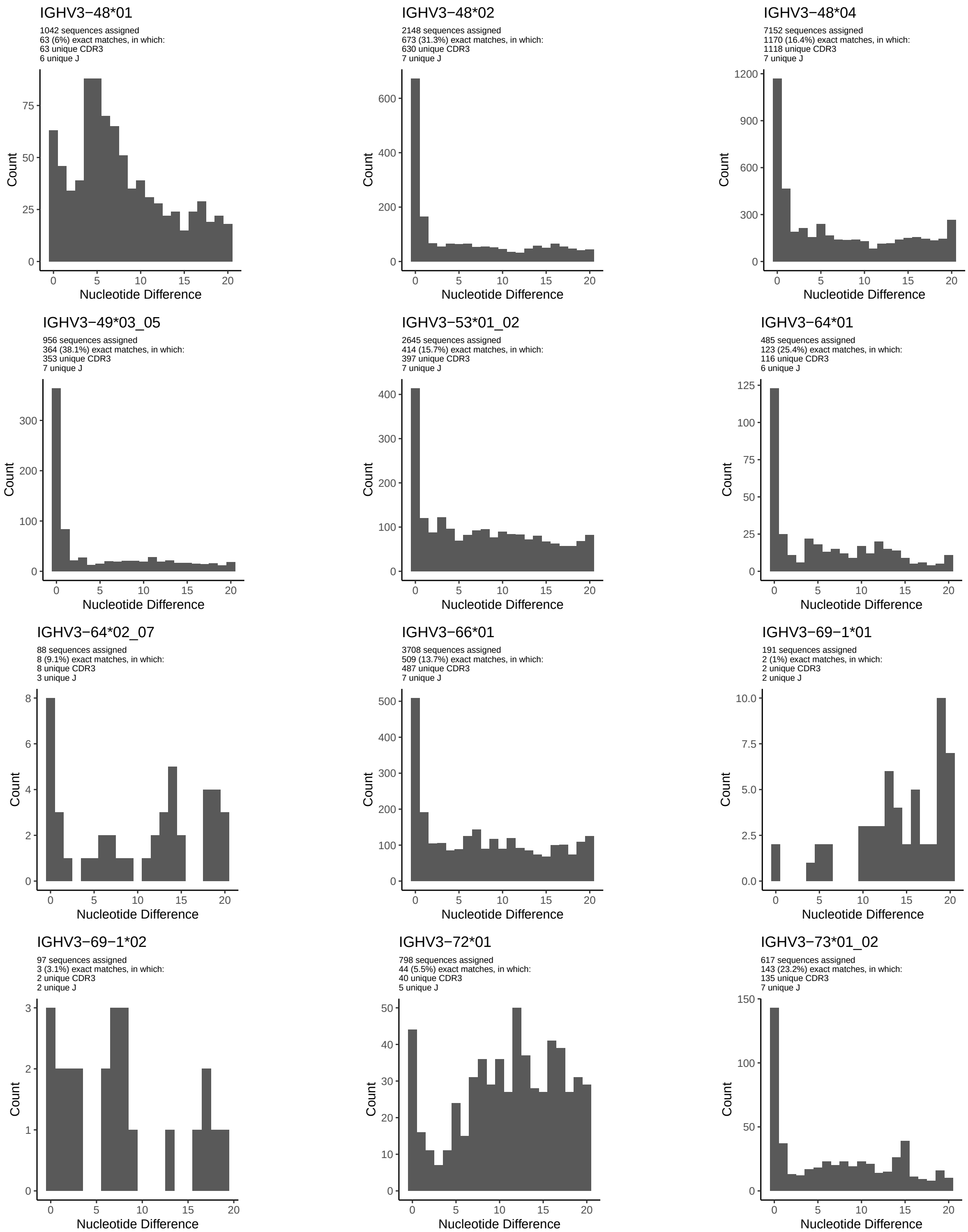
# 2 Variation from germline, in assignments to each allele

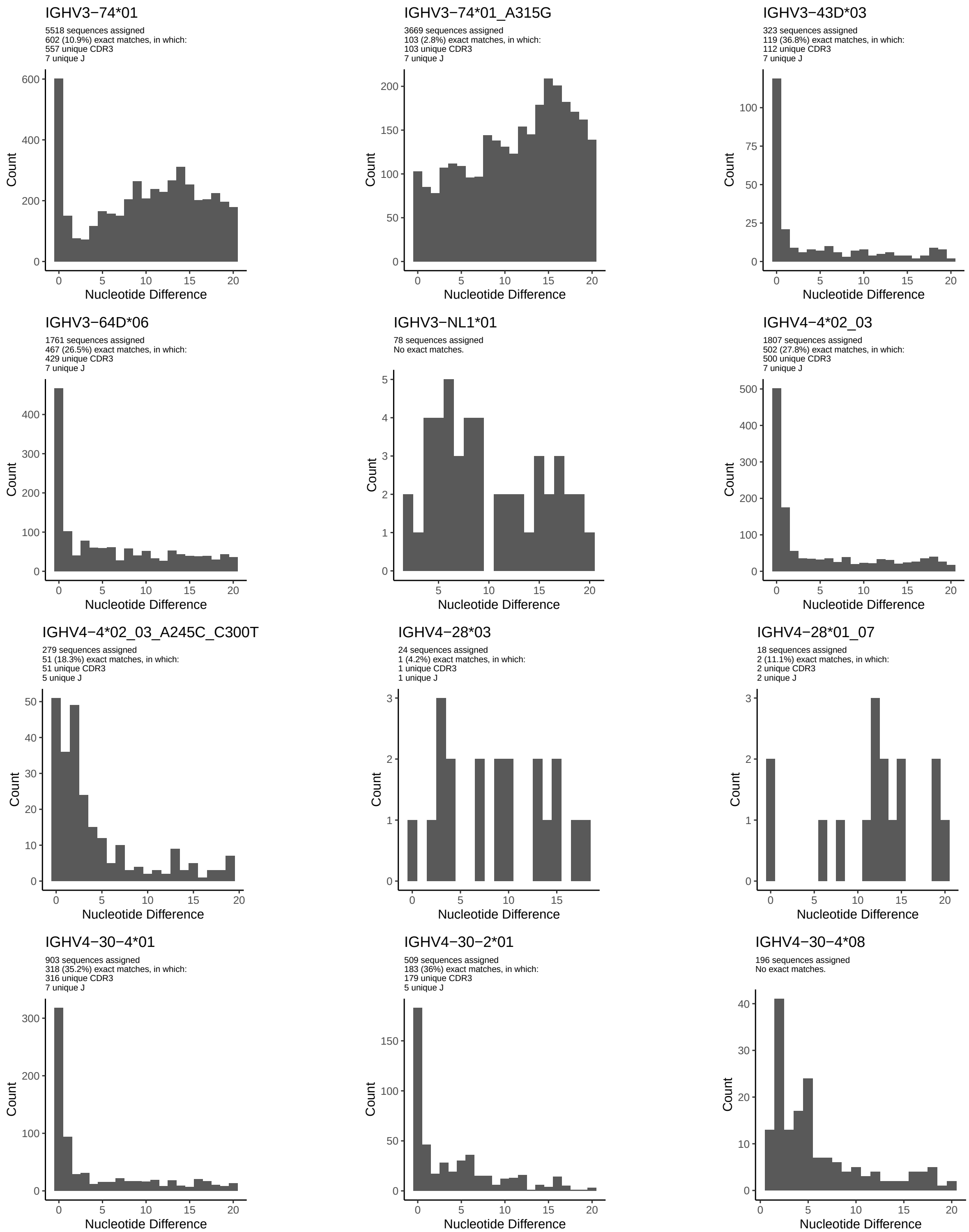


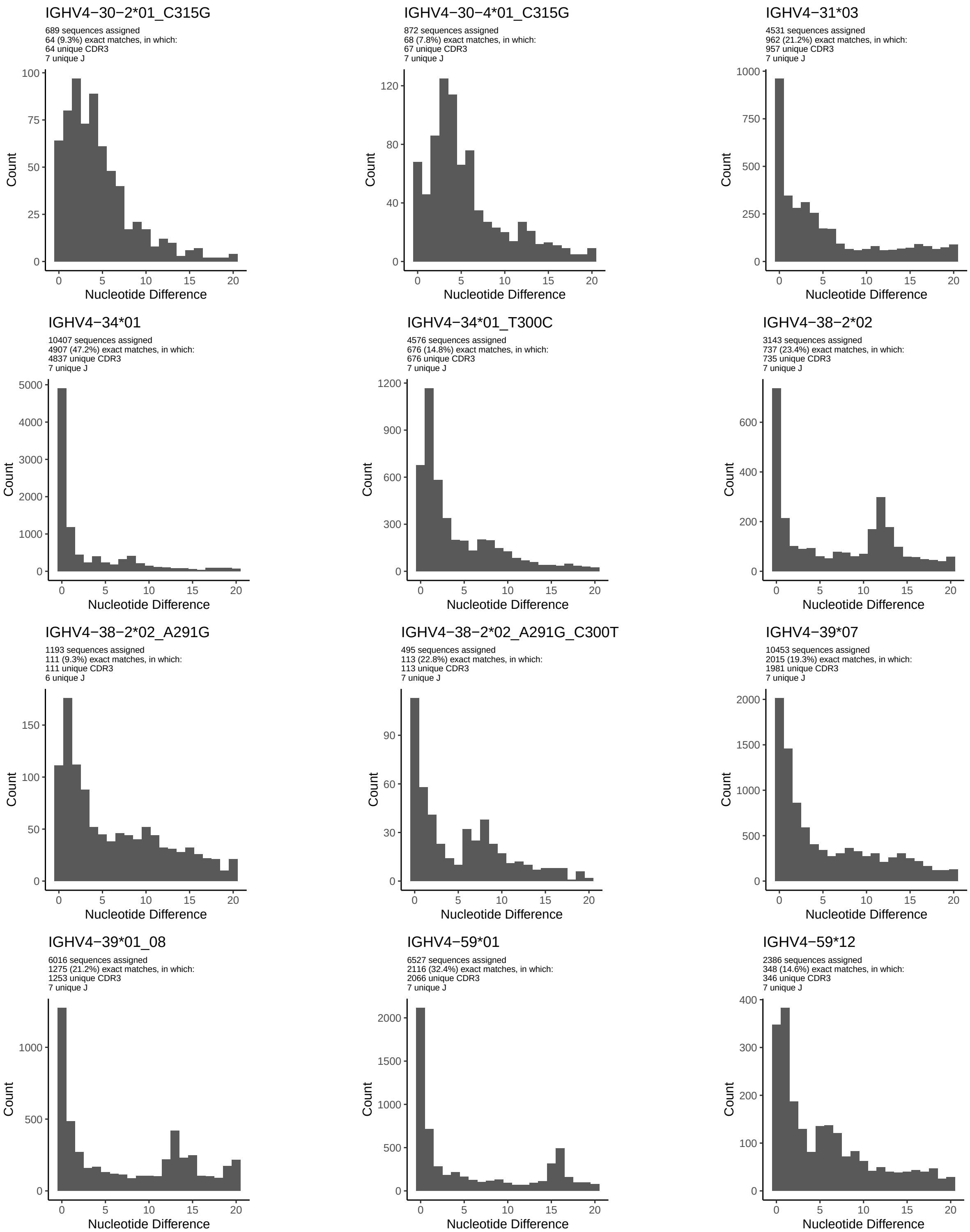

IGHV1−58*01_03
396 sequences assigned
184 (46.5%) exact matches, in which:
184 unique CDR3
6 unique J
IGHV1−69*01_12_13
6279 sequences assigned
2642 (42.1%) exact matches, in which:
2620 unique CDR3
7 unique J
IGHV1−69*04_09
4604 sequences assigned
1927 (41.9%) exact matches, in which:
1912 unique CDR3
7 unique J
IGHV2−5*02
2243 sequences assigned
788 (35.1%) exact matches, in which:
787 unique CDR3
7 unique J
IGHV2−26*01
6 sequences assigned
1 (16.7%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV2−26*02
15 sequences assigned
2 (13.3%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV2−26*04
528 sequences assigned
20 (3.8%) exact matches, in which:
20 unique CDR3
5 unique J
IGHV2−70*01
335 sequences assigned
107 (31.9%) exact matches, in which:
107 unique CDR3
5 unique J
IGHV3−7*01
6981 sequences assigned
951 (13.6%) exact matches, in which:
904 unique CDR3
7 unique J
IGHV3−7*03
748 sequences assigned
38 (5.1%) exact matches, in which:
38 unique CDR3
6 unique J
IGHV3−7*04
2801 sequences assigned
473 (16.9%) exact matches, in which:
443 unique CDR3
7 unique J
IGHV3−7*01_G144A
1153 sequences assigned
186 (16.1%) exact matches, in which:
184 unique CDR3
7 unique J
Count
Nucleotide Difference

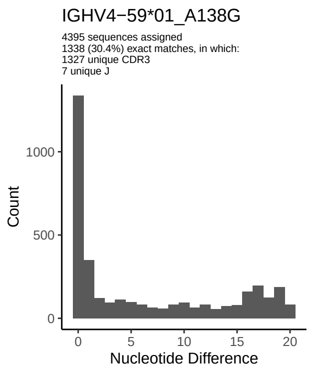

IGHV3−9*01
3626 sequences assigned
887 (24.5%) exact matches, in which:
831 unique CDR3
7 unique J
IGHV3−11*01
2308 sequences assigned
596 (25.8%) exact matches, in which:
564 unique CDR3
7 unique J
IGHV3−11*04
1014 sequences assigned
93 (9.2%) exact matches, in which:
93 unique CDR3
7 unique J
IGHV3−11*05
3741 sequences assigned
1026 (27.4%) exact matches, in which:
979 unique CDR3
7 unique J
IGHV3−11*06
1228 sequences assigned
210 (17.1%) exact matches, in which:
210 unique CDR3
7 unique J
IGHV3−13*01
480 sequences assigned
147 (30.6%) exact matches, in which:
135 unique CDR3
6 unique J
IGHV3−13*04
1329 sequences assigned
370 (27.8%) exact matches, in which:
344 unique CDR3
6 unique J
IGHV3−13*06
98 sequences assigned
1 (1%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV3−15*01
3526 sequences assigned
934 (26.5%) exact matches, in which:
902 unique CDR3
7 unique J
IGHV3−20*04
1377 sequences assigned
225 (16.3%) exact matches, in which:
217 unique CDR3
7 unique J
IGHV3−21*01
11405 sequences assigned
3445 (30.2%) exact matches, in which:
3238 unique CDR3
7 unique J
IGHV3−23*01_04
19825 sequences assigned
3067 (15.5%) exact matches, in which:
2868 unique CDR3
7 unique J
Count
Nucleotide Difference

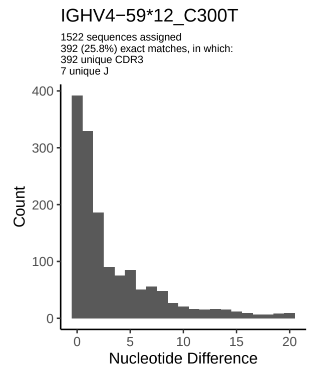

IGHV3−30−3*01
5356 sequences assigned
2181 (40.7%) exact matches, in which:
2082 unique CDR3
7 unique J
IGHV3−30*03
1660 sequences assigned
451 (27.2%) exact matches, in which:
445 unique CDR3
7 unique J
IGHV3−30*18
7275 sequences assigned
2162 (29.7%) exact matches, in which:
2054 unique CDR3
7 unique J
IGHV3−30−3*01_T288C
1777 sequences assigned
187 (10.5%) exact matches, in which:
186 unique CDR3
7 unique J
IGHV3−30*03_T288C
2089 sequences assigned
308 (14.7%) exact matches, in which:
306 unique CDR3
7 unique J
IGHV3−33*01
4848 sequences assigned
1167 (24.1%) exact matches, in which:
1123 unique CDR3
7 unique J
IGHV3−33*08
1289 sequences assigned
169 (13.1%) exact matches, in which:
168 unique CDR3
7 unique J
IGHV3−33*06_C288T
768 sequences assigned
165 (21.5%) exact matches, in which:
165 unique CDR3
7 unique J
IGHV3−35*01
8 sequences assigned
2 (25%) exact matches, in which:
2 unique CDR3
1 unique J
IGHV3−38−3*01
29 sequences assigned
6 (20.7%) exact matches, in which:
6 unique CDR3
3 unique J
IGHV3−38*03
4 sequences assigned
No exact matches.
IGHV3−43*01
898 sequences assigned
170 (18.9%) exact matches, in which:
163 unique CDR3
7 unique J
Count
Nucleotide Difference

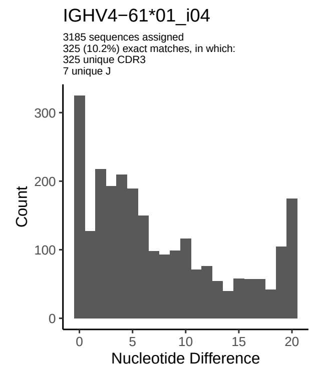

IGHV3−48*01
1042 sequences assigned
63 (6%) exact matches, in which:
63 unique CDR3
6 unique J
IGHV3−48*02
2148 sequences assigned
673 (31.3%) exact matches, in which:
630 unique CDR3
7 unique J
IGHV3−48*04
7152 sequences assigned
1170 (16.4%) exact matches, in which:
1118 unique CDR3
7 unique J
IGHV3−49*03_05
956 sequences assigned
364 (38.1%) exact matches, in which:
353 unique CDR3
7 unique J
IGHV3−53*01_02
2645 sequences assigned
414 (15.7%) exact matches, in which:
397 unique CDR3
7 unique J
IGHV3−64*01
485 sequences assigned
123 (25.4%) exact matches, in which:
116 unique CDR3
6 unique J
IGHV3−64*02_07
88 sequences assigned
8 (9.1%) exact matches, in which:
8 unique CDR3
3 unique J
IGHV3−66*01
3708 sequences assigned
509 (13.7%) exact matches, in which:
487 unique CDR3
7 unique J
IGHV3−69−1*01
191 sequences assigned
2 (1%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−69−1*02
97 sequences assigned
3 (3.1%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV3−72*01
798 sequences assigned
44 (5.5%) exact matches, in which:
40 unique CDR3
5 unique J
IGHV3−73*01_02
617 sequences assigned
143 (23.2%) exact matches, in which:
135 unique CDR3
7 unique J
Count
Nucleotide Difference

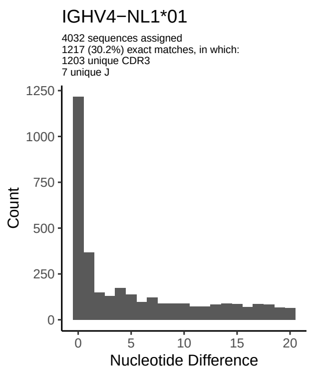

IGHV3−74*01
5518 sequences assigned
602 (10.9%) exact matches, in which:
557 unique CDR3
7 unique J
IGHV3−74*01_A315G
3669 sequences assigned
103 (2.8%) exact matches, in which:
103 unique CDR3
7 unique J
IGHV3−43D*03
323 sequences assigned
119 (36.8%) exact matches, in which:
112 unique CDR3
7 unique J
IGHV3−64D*06
1761 sequences assigned
467 (26.5%) exact matches, in which:
429 unique CDR3
7 unique J
IGHV3−NL1*01
78 sequences assigned
No exact matches.
IGHV4−4*02_03
1807 sequences assigned
502 (27.8%) exact matches, in which:
500 unique CDR3
7 unique J
IGHV4−4*02_03_A245C_C300T
279 sequences assigned
51 (18.3%) exact matches, in which:
51 unique CDR3
5 unique J
IGHV4−28*03
24 sequences assigned
1 (4.2%) exact matches, in which:
1 unique CDR3
1 unique J
IGHV4−28*01_07
18 sequences assigned
2 (11.1%) exact matches, in which:
2 unique CDR3
2 unique J
IGHV4−30−4*01
903 sequences assigned
318 (35.2%) exact matches, in which:
316 unique CDR3
7 unique J
IGHV4−30−2*01
509 sequences assigned
183 (36%) exact matches, in which:
179 unique CDR3
5 unique J
IGHV4−30−4*08
196 sequences assigned
No exact matches.
Count
Nucleotide Difference

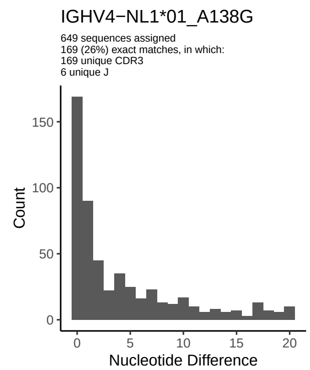

IGHV4−30−2*01_C315G
689 sequences assigned
64 (9.3%) exact matches, in which:
64 unique CDR3
7 unique J
IGHV4−30−4*01_C315G
872 sequences assigned
68 (7.8%) exact matches, in which:
67 unique CDR3
7 unique J
IGHV4−31*03
4531 sequences assigned
962 (21.2%) exact matches, in which:
957 unique CDR3
7 unique J
IGHV4−34*01
10407 sequences assigned
4907 (47.2%) exact matches, in which:
4837 unique CDR3
7 unique J
IGHV4−34*01_T300C
4576 sequences assigned
676 (14.8%) exact matches, in which:
676 unique CDR3
7 unique J
IGHV4−38−2*02
3143 sequences assigned
737 (23.4%) exact matches, in which:
735 unique CDR3
7 unique J
IGHV4−38−2*02_A291G
1193 sequences assigned
111 (9.3%) exact matches, in which:
111 unique CDR3
6 unique J
IGHV4−38−2*02_A291G_C300T
495 sequences assigned
113 (22.8%) exact matches, in which:
113 unique CDR3
7 unique J
IGHV4−39*07
10453 sequences assigned
2015 (19.3%) exact matches, in which:
1981 unique CDR3
7 unique J
IGHV4−39*01_08
6016 sequences assigned
1275 (21.2%) exact matches, in which:
1253 unique CDR3
7 unique J
IGHV4−59*01
6527 sequences assigned
2116 (32.4%) exact matches, in which:
2066 unique CDR3
7 unique J
IGHV4−59*12
2386 sequences assigned
348 (14.6%) exact matches, in which:
346 unique CDR3
7 unique J
Count
Nucleotide Difference

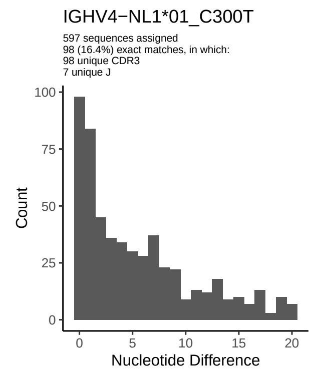

IGHV4−59*01_A138G
4395 sequences assigned
1338 (30.4%) exact matches, in which:
1327 unique CDR3
7 unique J
Count
Nucleotide Difference

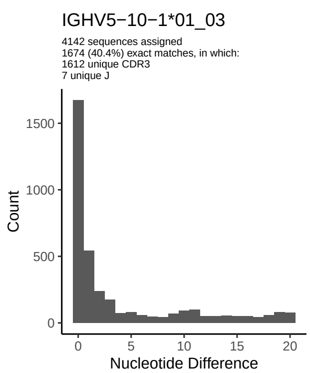

IGHV4−59*12_C300T
1522 sequences assigned
392 (25.8%) exact matches, in which:
392 unique CDR3
7 unique J
Count
Nucleotide Difference

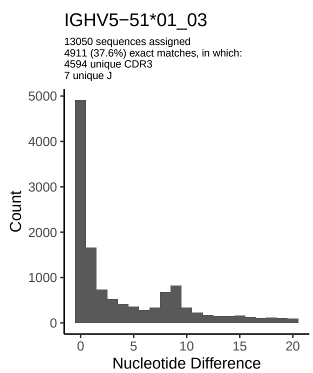

IGHV4−61*01_i04
3185 sequences assigned
325 (10.2%) exact matches, in which:
325 unique CDR3
7 unique J
Count
Nucleotide Difference

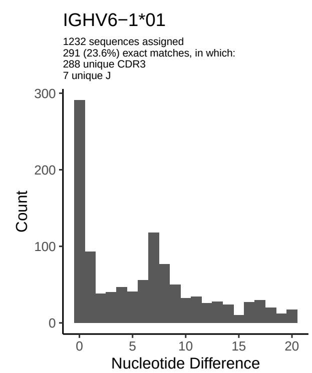

IGHV4−NL1*01
4032 sequences assigned
1217 (30.2%) exact matches, in which:
1203 unique CDR3
7 unique J
Count
Nucleotide Difference

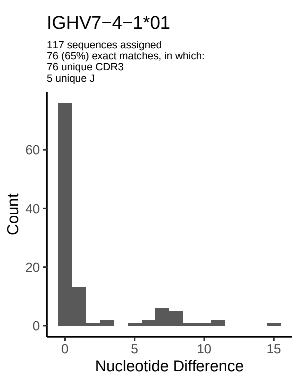

IGHV4−NL1*01_A138G
649 sequences assigned
169 (26%) exact matches, in which:
169 unique CDR3
6 unique J
Count
Nucleotide Difference


IGHV4−NL1*01_C300T
597 sequences assigned
98 (16.4%) exact matches, in which:
98 unique CDR3
7 unique J
Count
Nucleotide Difference

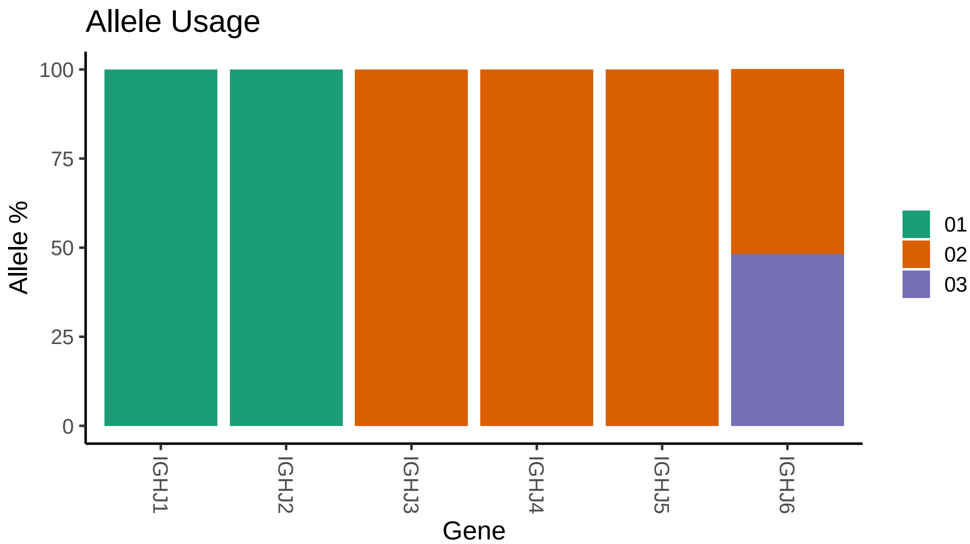

IGHV5−10−1*01_03
4142 sequences assigned
1674 (40.4%) exact matches, in which:
1612 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV5−51*01_03
13050 sequences assigned
4911 (37.6%) exact matches, in which:
4594 unique CDR3
7 unique J
Count
Nucleotide Difference

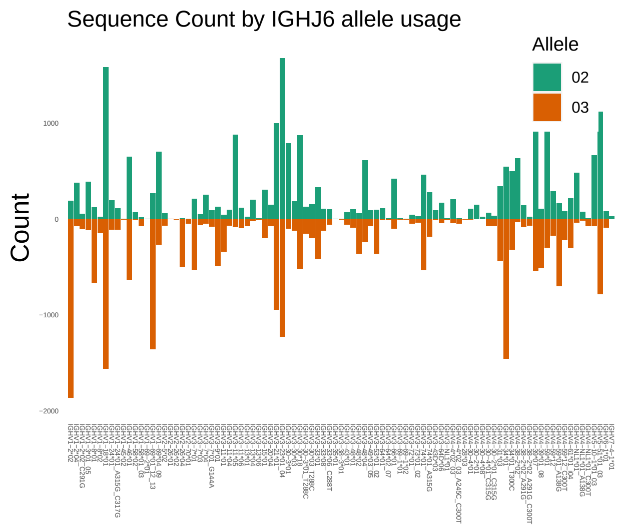

IGHV6−1*01
1232 sequences assigned
291 (23.6%) exact matches, in which:
288 unique CDR3
7 unique J
Count
Nucleotide Difference


IGHV7−4−1*01
117 sequences assigned
76 (65%) exact matches, in which:
76 unique CDR3
5 unique J
Count
Nucleotide Difference

## 3 Allele usage in potential haplotype anchor genes



# 4 Haplotype plots



# 5 Configuration settings

```
## Repertoire file: /misc/work/jenkins/PRJNA324093/Donor 157/036 S024/157_IGH/157_IGH/a2/7e65d7c7ff647f8
##
## Germline reference file: /misc/work/jenkins/PRJNA324093/Donor 157/036 S024/157_IGH/157_IGH/a2/7e65d7c
##
## Novel allele file: /misc/work/jenkins/PRJNA324093/Donor 157/036 S024/157_IGH/157_IGH/a2/7e65d7c7ff647
##
## Species: Homosapiens
##
## Chain: IGHV
##
## Segment: V
##
## Unexpected N in reference sequence IGHV1 2 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 2 02_C291G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV1 24 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV1 24 01_A315G_C317G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 7 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 7 01_G144A: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 03_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 30 3 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 30 3 01_T288C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 33 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 33 06_C288T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV3 74 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV3 74 01_A315G: replacing with gap
```

```
##
##
## Unexpected N in reference sequence IGHV4 4 02_03: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 4 02_03_A245C_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 2 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 2 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 30 4 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 30 4 01_C315G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 34 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 34 01_T300C: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 38 2 02: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 38 2 02_A291G_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 01_A138G: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 59 12: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 59 12_C300T: replacing with gap
##
##
## Unexpected N in reference sequence IGHV4 NL1 01: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_A138G: replacing with gap
##
##
## Unexpected N in novel sequence IGHV4 NL1 01_C300T: replacing with gap
```